Microsoft Elevate

CAPSTONE PROJECT

Hospital Management

PRESENTED BY

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OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result
- Conclusion
- Future Scope
- References

PROBLEM STATEMENT:

Hospitals face several operational challenges that make it difficult for administrators to effectively manage services and improve patient care. One major issue is the difficulty in tracking patient volume across different departments, which makes it hard to allocate resources efficiently. In addition, there is often a lack of visibility into patient waiting times, preventing staff from identifying delays and improving service speed. Hospitals also struggle with poor monitoring of department referral patterns, making it challenging to understand how patients move through the healthcare system. Analyzing patient satisfaction levels is another difficulty, as the available data is often scattered or not properly organized. Furthermore, many hospitals still rely on manual reports, which are time-consuming to prepare and prone to human errors. Without proper data visualization and analytics tools, hospital administrators find it difficult to identify operational inefficiencies and implement improvements that could enhance overall patient services..

PROPOSED SOLUTION:

The proposed system aims to improve hospital management by analyzing patient data such as visits, department referrals, waiting time, and satisfaction scores. Using **data analytics and Power BI visualization**, the system provides clear insights into hospital performance and patient flow.

Data Collection

Hospital patient data is collected from healthcare records, including patient name, age, gender, race, department referral, waiting time, satisfaction score, month, and year.

Data Preprocessing

The collected data is cleaned and prepared by:

Handling missing values,

Removing duplicate records

Correcting inconsistent formats

Converting data types for analysis

PROPOSED SOLUTION:

Data Visualization

Interactive dashboards are created in Power BI using:

Bar charts for department referrals

Pie charts for gender distribution

Line charts for monthly patient trends

Filters for department, age, month, and year

Data Modeling and Analysis

Power BI data modeling is used to organize the dataset and calculate important metrics such as:

Total number of patients

Average waiting time

Patient satisfaction score

Department referral counts

This system helps healthcare management **analyze patient information, improve decision-making, and enhance hospital services.**

SYSTEM APPROACH:

The system approach for the Hospital Management Dashboard follows a step-by-step process to convert raw hospital data into meaningful insights.

- **Data Collection** – Hospital patient data such as age, gender, department referral, waiting time, and satisfaction score is collected.
- **Data Preparation** – The data is cleaned and organized by removing duplicates, handling missing values, and correcting formats.
- **Data Analysis** – Power BI is used to calculate key metrics such as total patients, average waiting time, and department referral counts.
- **Dashboard Development** – Interactive visualizations such as charts, KPI cards, and graphs are created to represent hospital data.
- **Insight Generation** – The dashboard helps administrators monitor hospital performance and make data-driven decisions.

ALGORITHM & DEPLOYMENT:

Algorithm / Techniques Used:

- **Data Aggregation:** Aggregation techniques are used to summarize hospital data, such as calculating the **total number of patients, department referrals, and case counts**.
- **Average Calculation:** Average functions are used to compute **average patient waiting time**, helping to analyze hospital service efficiency.
- **Grouping and Filtering:** Data is grouped based on **department, gender, month, and year** to identify patterns in patient visits and referrals.
- **Trend Analysis:** Line charts are used to analyze **monthly and yearly patient trends**, helping understand variations in hospital visits.

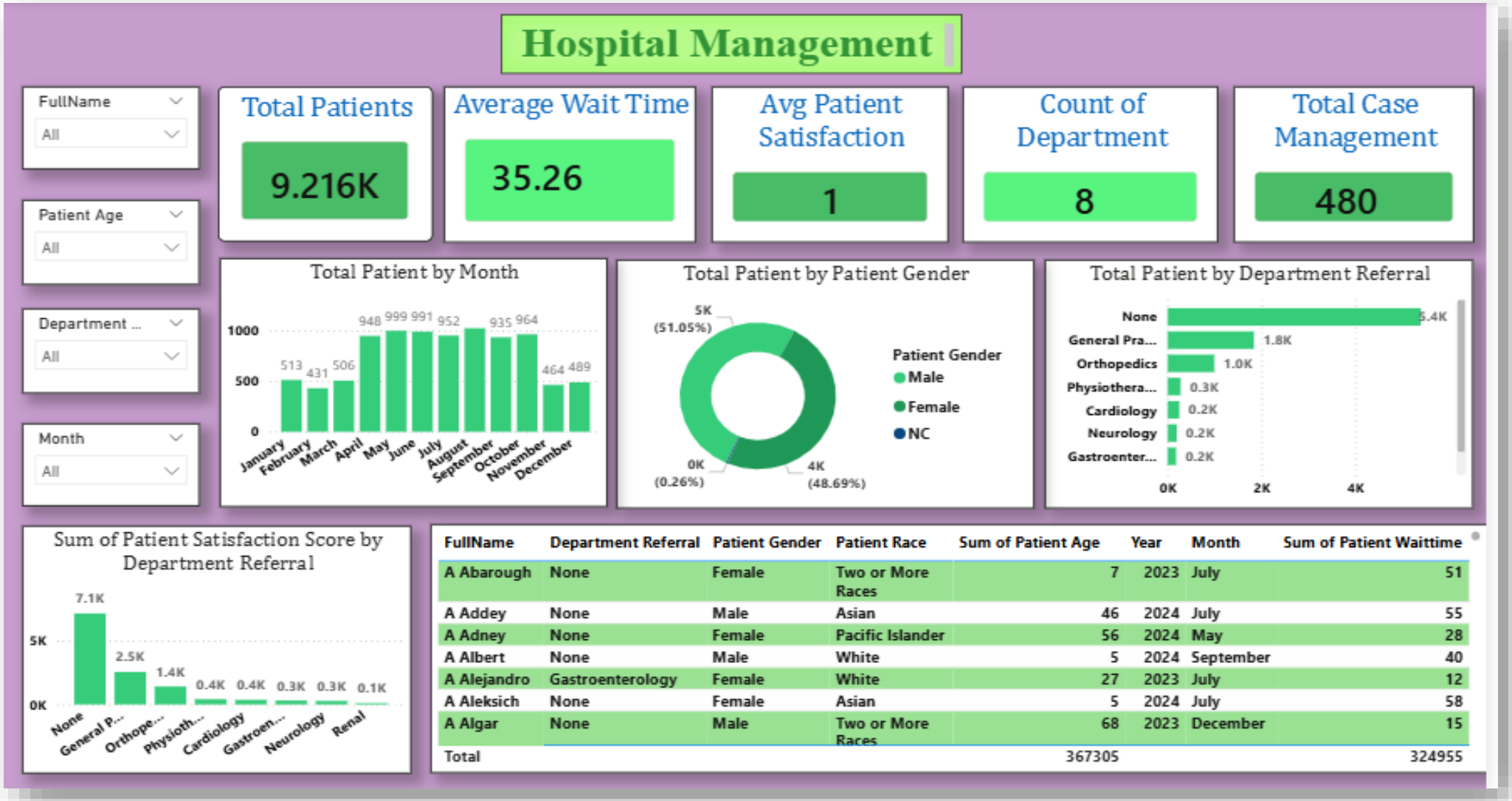
Deployment

- The dashboard is developed using **Power BI Desktop** and deployed through **Power BI Service**.

Steps involved in deployment:

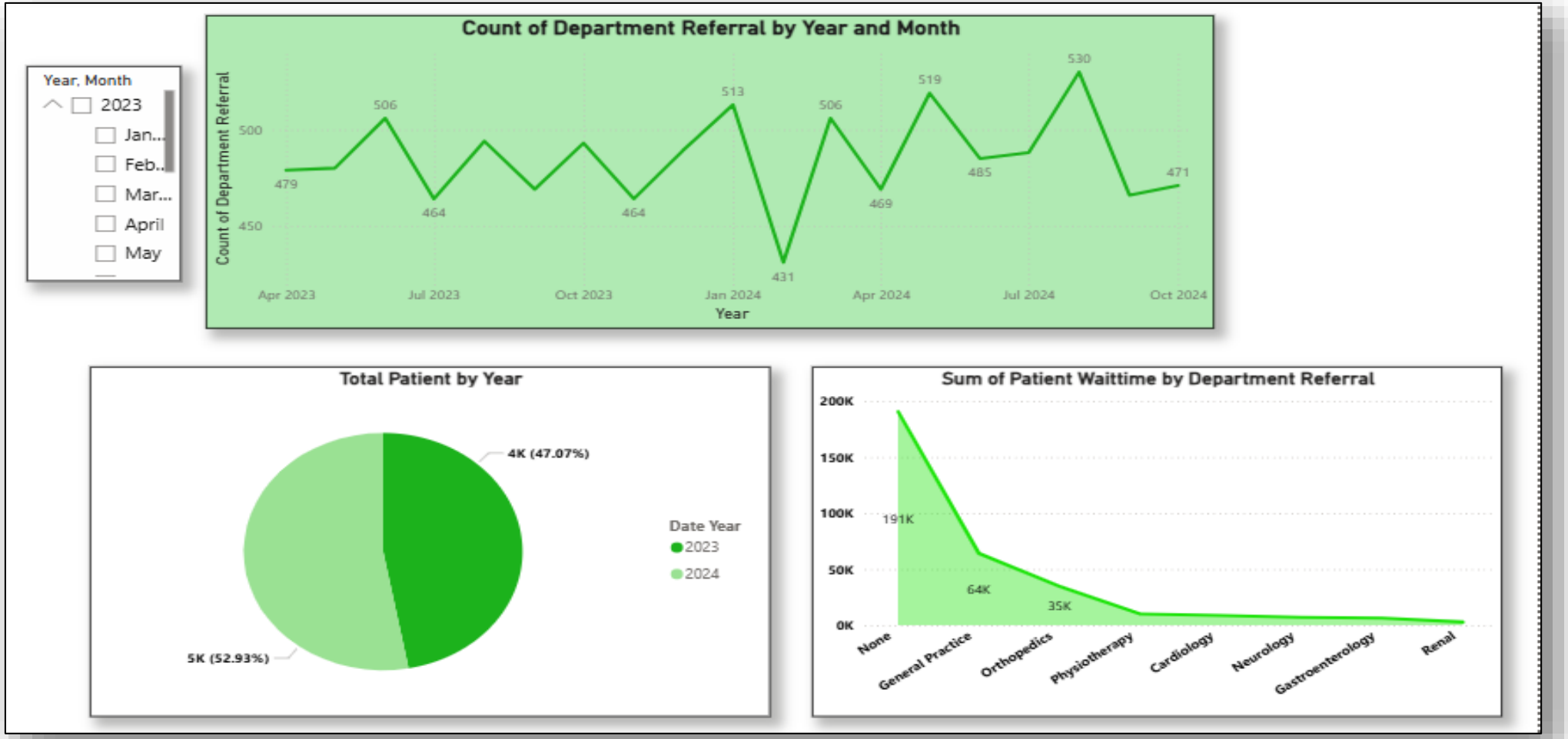
- Import and prepare hospital dataset in Power BI.
- Design interactive dashboards with charts and KPI indicators.
- Publish the dashboard to **Power BI Service**.
- Share the dashboard with hospital administrators for monitoring and analysis.
- The deployed dashboard allows users to **access hospital data insights in real time and support better decision-making.**

RESULT:



FullName	Department Referral	Patient Gender	Patient Race	Sum of Patient Age	Year	Month	Sum of Patient Waittime
A Abarough	None	Female	Two or More Races	7	2023	July	51
A Addey	None	Male	Asian	46	2024	July	55
A Adney	None	Female	Pacific Islander	56	2024	May	28
A Albert	None	Male	White	5	2024	September	40
A Alejandro	Gastroenterology	Female	White	27	2023	July	12
A Aleksich	None	Female	Asian	5	2024	July	58
A Algar	None	Male	Two or More Races	68	2023	December	15
Total				367305			324955

RESULT:


Year, Month
2023
Jan...
Feb...
Mar...
April
May
Count of Department Referral by Year and Month
Count of Department Referral
500
450
479
506
464
464
513
431
506
469
519
485
530
471
Apr 2023
Jul 2023
Oct 2023
Jan 2024
Apr 2024
Jul 2024
Oct 2024
Year
Total Patient by Year
4K (47.07%)
5K (52.93%)
Date Year
2023
2024
Sum of Patient Waittime by Department Referral
200K
150K
100K
50K
0K
191K
64K
35K
None
General Practice
Orthopedics
Physiotherapy
Cardiology
Neurology
Gastroenterology
Renal

CONCLUSION:

- The Hospital Management Dashboard was successfully developed using **Power BI** to analyze hospital patient data.
- The dashboard provides insights into **patient visits, department referrals, waiting time, and satisfaction levels.**
- Interactive visualizations help hospital administrators **monitor performance and identify trends easily**.
- This system supports **data-driven decision making and improves hospital management efficiency**.

FUTURE SCOPE:

- Integrate the dashboard with **real-time hospital databases** to monitor patient data instantly.
- Use **machine learning models** to predict patient visits and waiting times.
- Add analysis for **doctor performance and hospital resource utilization**.
- Include **bed availability and appointment management tracking**.
- Develop **mobile or web-based access** for easier monitoring by hospital administrators.

REFERENCES:

GitHub Link: https://github.com/KancharlaNithyasri/Hospital-Management-Project-PowerBI.git

Thank You